

Compiled Details of Projects

1. Course Registration System – IIIT Vadodara

Project Title

Course Registration System – IIIT Vadodara

Supervisor

Dr. Pramit Majumdar

Project Abstract

A comprehensive web-based course registration system for IIIT Vadodara built using Django. The system allows students to register for courses, faculty members to review registrations, and administrators to manage the registration process. The project includes student and faculty dashboards, course filtering based on semester and branch, registration status tracking, email notifications, and profile management.

Objectives

- Build a centralized course registration platform for students and faculty.
- Provide secure login and profile management functionality.
- Enable dynamic course filtering based on branch and semester.
- Allow faculty members to approve or reject course registrations with feedback.
- Provide registration tracking and email notification support.
- Maintain academic schedules such as bus, mess, and class schedules.

Outcomes

- Developed a session-based web application using Django.
- Implemented student and faculty workflows for registration and approval.
- Added real-time course availability checking.
- Integrated email notifications for registration updates.
- Structured the project with modular Django applications including authentication, registration logic, templates, and static assets.

Technologies Used

- Django 5

- SQLite
- Bootstrap 4
- jQuery
- Django Forms
- Python

Code Repository

[Course Registration Repository](#)

Associated Achievements

- Built a complete academic workflow prototype for institutional course registration.
 - Implemented faculty review and approval mechanisms.
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2. Metaverse Project – Prompting 3D Assets with LLM Assistance

Project Title

Metaverse Project – Prompting 3D Assets with LLM Assistance

Supervisor

Dr. Pramit Majumdar

Project Abstract

A Unity-based project for interactive 3D object inspection integrated with a chat-style LLM assistant. The prototype allows users to click on objects in a 3D scene, analyze geometry, compare whether objects fit inside each other, inspect material and physics properties, generate transparent wrapping boxes, and interact using natural-language prompts. The project combines Unity, C#, TextMesh Pro, and OpenRouter-backed LLM integration for intent-based interaction.

Objectives

- Create an interactive 3D environment with object inspection capabilities.
- Enable natural-language interaction with 3D assets using LLMs.
- Support geometry analysis including dimensions, surface area, and volume.
- Implement object-to-object fit analysis.

- Provide scene object listing and property inspection.
- Integrate OpenRouter APIs for chat-based intent parsing and responses.

Outcomes

- Developed a Unity prototype with object selection and highlighting.
- Implemented geometry analysis utilities and fit-checking systems.
- Built a chat interface using TextMesh Pro.
- Added OpenRouter-backed LLM communication for structured intent generation.
- Integrated runtime object analysis including volume, dimensions, material information, and wrap-box generation.
- Implemented scripts for object selection, camera movement, auto-scroll chat, and scene object listing.

Technologies Used

- Unity 2022.3.62f3
- C#
- TextMesh Pro
- Unity UI (uGUI)
- OpenRouter Chat Completions API
- Unity Visual Scripting

Code Repository

[Prompting 3D Assets Repository](#)

Associated Achievements

- Integrated LLM-driven interaction into a 3D Unity environment.
 - Enabled natural-language geometry and scene analysis.
 - Built a prototype combining metaverse interaction with AI-assisted object analysis.
 - Implemented runtime object wrapping and fit visualization features.
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3. RDMA File Transfer Demo

Project Title

RDMA File Transfer Demo

Supervisor

Dr. Bhupendra Kumar

Project Abstract

A GUI-based application demonstrating the differences between TCP and RDMA file transfer protocols. The application provides a modern dark-themed interface with file selection, TCP transfer simulation, RDMA transfer simulation, and real-time transfer status logging to help visualize and compare transfer mechanisms.

Objectives

- Demonstrate differences between TCP and RDMA file transfer protocols.
- Build a graphical interface for transfer simulations.
- Provide real-time transfer logging and status updates.
- Visualize protocol behavior using interactive UI components.
- Create a modern user-friendly desktop application.

Outcomes

- Developed a Python-based GUI application.
- Implemented TCP transfer simulation.
- Implemented RDMA transfer simulation.
- Added real-time transfer status logging.
- Designed a modern dark-themed interface with hover effects and smooth animations.

Technologies Used

- Python 3.6+
- Tkinter

Code Repository

[RDMA File Transfer](#)

Associated Achievements

- Built a desktop-based visualization tool for networking concepts.
- Demonstrated RDMA and TCP transfer workflow simulation through an interactive GUI.
- Implemented modern UI interaction features in a Python desktop application.